Task 1: Linear Regression Model

House Price Predictor — California Housing Dataset

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1. Objective

This report summarizes the exploratory data analysis, preprocessing, model training, and evaluation performed for a Linear Regression model that predicts median house value using the California Housing dataset (20,640 census block groups).

2. Dataset & Exploratory Data Analysis

The dataset contains 9 numeric features (location, housing age, rooms, bedrooms, population, households, median income) plus one categorical feature, ocean_proximity. The target variable, median house value, ranges from about $14,999 to $500,001 (capped). The total_bedrooms column had 207 missing values (about 1% of rows), which were imputed with the column median. The categorical ocean_proximity feature was one-hot encoded before modeling.

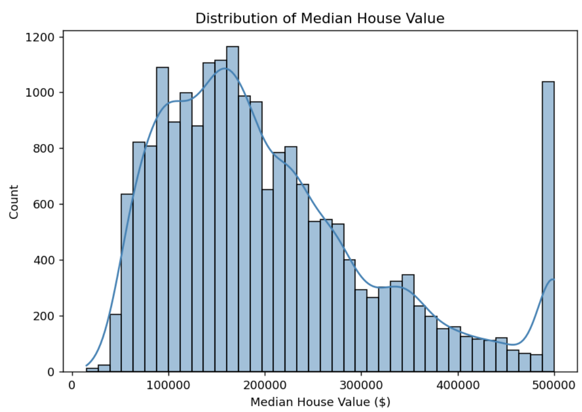

Figure 1: Distribution of median house value — right-skewed, with a visible cap near $500,000.

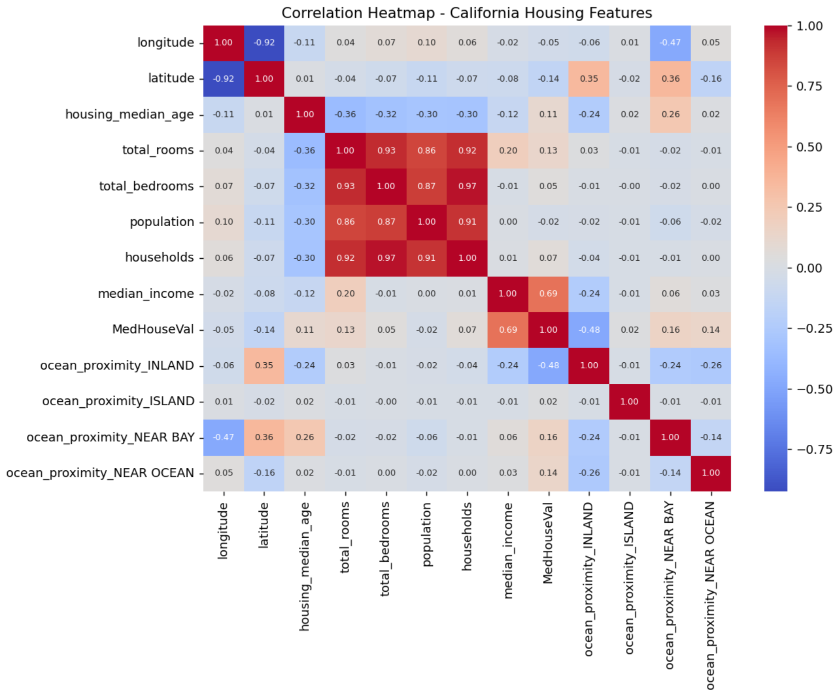

Figure 2: Correlation heatmap. median_income shows the strongest positive correlation with house value.

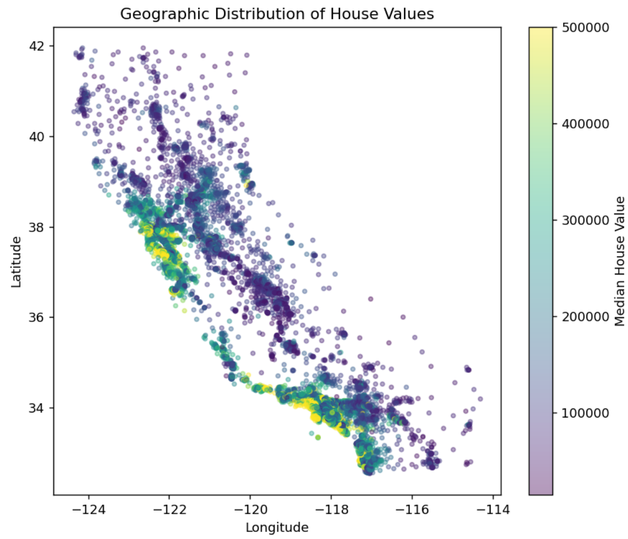

Figure 3: Geographic distribution — coastal areas (esp. near San Francisco and Los Angeles) show higher values.

3. Model & Evaluation

An 80/20 train/test split (random_state=42) was used with scikit-learn's LinearRegression on all numeric + one-hot-encoded features. Performance on the held-out test set:

Metric	Value	Meaning
MAE	$50,671	Average absolute prediction error
RMSE	$70,061	Root mean squared error (penalizes large errors)
R^2	0.625	~62.5% of variance in house value explained

The strongest positive coefficients came from ocean_proximity = ISLAND and median_income; the strongest negative coefficient was ocean_proximity = INLAND, meaning inland homes tend to be cheaper, all else equal.

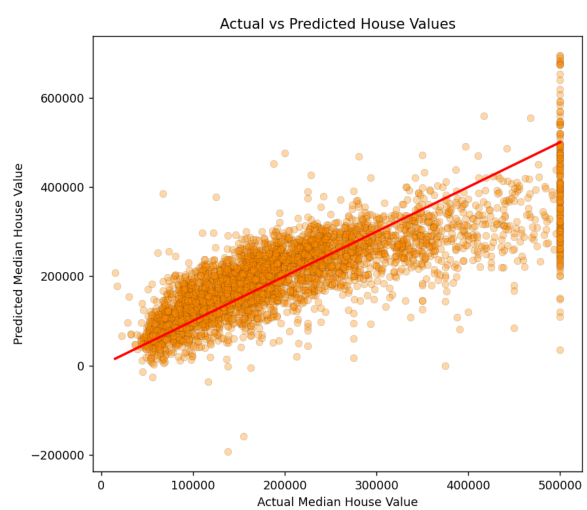

Figure 4: Actual vs predicted values. Points near the red line are accurate; the model underpredicts very high-value homes.

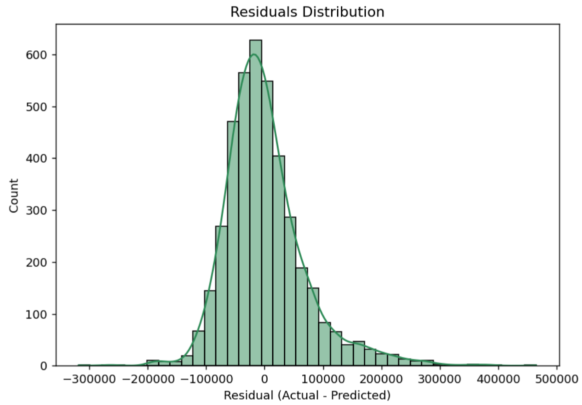

Figure 5: Residuals are roughly centered at zero but slightly right-skewed, indicating the linear model misses some higher-value homes.

4. Conclusion & Improvement Ideas

The linear regression model provides a reasonable baseline ($R^2 \approx 0.62$) but leaves noticeable error, especially at the high end of house values. Suggested next steps:

• Try non-linear models such as Random Forest or Gradient Boosting (XGBoost/LightGBM) for likely accuracy gains.
• Engineer new features: rooms per household, bedrooms per room, population per household.
• Apply feature scaling and regularization (Ridge/Lasso) to stabilize coefficients.
• Address the value cap at $500,001, which biases predictions for expensive homes.
• Use k-fold cross-validation for a more robust performance estimate.

Deliverables

task1_ml_linear_regression.ipynb (full code, EDA, plots, comments), this PDF report, and model.pkl (saved trained model).